MOHAMED ELSAID

(914) 334-1436 | melsaid2017@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Pharmaceutical scientist with an M.S. in Bioinformatics (Johns Hopkins, 2026) and 3+ years of industry experience at Pfizer and Catalent. Develops scientific data workflows in Python, R, SQL, and Bash to process NGS and biomedical datasets, with deep expertise in RNA-seq analysis, variant calling, machine learning, and GMP/GLP data integrity. Seeking to bridge pharmaceutical rigor and computational biology skills in a scientific data engineering role at NIEHS.

TECHNICAL SKILLS

Programming: Python, R, SQL, Bash, Linux/Unix, Git

Bioinformatics: RNA-seq, NGS Pipelines, Variant Calling, STAR, GATK, SAMtools, FastQC, DESeq2, GO/KEGG

Data Science: scikit-learn, Random Forest, SVM, Statistical Modeling, Data Visualization

Laboratory: GMP, GLP, LIMS, ALCOA+, Data Integrity, 21 CFR Part 11, SOP Development

PROFESSIONAL EXPERIENCE

Associate Scientist, QC Microbiology | Catalent Pharma Solutions — Harmans, MD *Apr 2025 – Oct 2025*

- Executed GMP analytical workflows processing 200+ samples per week for sterility and bioburden testing; applied measurement uncertainty quantification to identify out-of-specification trends and support batch disposition.
- Enforced ALCOA+ data integrity standards across the full sample lifecycle via LIMS-based workflows and audit trail review, achieving zero data integrity findings over a 6-month GMP period.
- Authored and revised 10+ SOPs and technical reports compliant with 21 CFR Part 11; led a documentation redesign adopted department-wide that improved audit readiness and reduced review cycles.
- Coordinated cross-functional response among Manufacturing, QA, and R&D to resolve instrumentation failures and deliver analytical results under time-critical batch release constraints.

Associate Scientist | Pfizer Inc. — Pearl River, NY *Aug 2022 – Feb 2024*

- Supported high-throughput immunogenicity testing for SARS-CoV-2 vaccine programs, executing quantitative data collection and assay analysis across 800–1,000 samples per week over 18 months of active vaccine development.
- Performed statistical analysis and variability assessments across 15+ large-scale biomedical datasets using R; produced analytical data packages directly supporting FDA and international regulatory submissions.
- Redesigned sample processing workflows with calibration controls and quality checkpoints, reducing analytical turnaround time by ~15% and eliminating recurring manual bottlenecks.
- Authored 20+ GMP-grade SOPs, method validations, and analytical reports supporting clinical operations, QA, and laboratory management stakeholders.

TECHNICAL PROJECTS

RNA-seq Analysis Pipeline | JHU: Bioinformatics Tools for Genome Analysis *2025*

- Built an end-to-end RNA-seq workflow (FastQC, Trimmomatic, STAR, featureCounts, DESeq2) identifying differentially expressed genes across 3 replicate sets with GO/KEGG pathway enrichment; version-controlled in GitHub for full reproducibility.

ML Classification Pipeline | JHU: Applied Machine Learning *2026*

- Developed and benchmarked random forest, logistic regression, and SVM classifiers using scikit-learn on a biological dataset; optimized via 5-fold CV and feature engineering; reported ROC/AUC, confusion matrices, and feature importance.

GC Content Web Application | JHU: Software Engineering for Bioinformatics *2025*

- Built a Python web app for FASTA sequence analysis with sliding-window GC calculations, SQL database integration, Linux deployment, and Git version control. demonstrating full-stack scientific computing skills.

EDUCATION

M.S. in Bioinformatics — Johns Hopkins University, Baltimore, MD *Sep 2024 – May 2026*

B.S. in Biochemistry — Manhattan College, Bronx, NY *Sep 2018 – May 2022*

Note: Oct 2025 – May 2026 dedicated to full-time M.S. completion at Johns Hopkins University.